

Code:

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#include <stdio.h>
#include <string.h>
#include <stdlib.h>

#define CMDS 5
#define TOKENS 4
#define MAXWORDS 50
#define LEN 256

char *tokens[TOKENS] = {"[N]", "[AV]", "[V]", "[AJ]"};
char *cmds[CMDS] = {"NOUNS", "ADVERBS", "VERBS", "ADJECTIVES", "END"};

char *lists[CMDS][MAXWORDS];
int cl[CMDS];
int used[CMDS];

int main() {
    char template_str[LEN];
    if (!fgets(template_str, sizeof(template_str), stdin)) return 0;
    template_str[strcspn(template_str, "\r\n")] = '\0';

    char line[LEN];
    int current_cmd = -1;

    while (fgets(line, sizeof(line), stdin)) {
        line[strcspn(line, "\r\n")] = '\0';
        if (strcmp(line, "END") == 0) break;

        int found_cmd = -1;
        for (int i = 0; i < CMDS - 1; i++) {
            if (strcmp(line, cmds[i]) == 0) {
                found_cmd = i;
                break;
            }
        }

        if (found_cmd != -1) {
            current_cmd = found_cmd;
        } else if (current_cmd != -1 && strlen(line) > 0) {
            lists[current_cmd][cl[current_cmd]] = strdup(line);
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        cl[current_cmd]++;
    }
}

for (int pass = 0; pass < 2; pass++) {
    char buffer[1024];
    buffer[0] = '\0';
    char copy[LEN];
    strcpy(copy, template_str);

    char *token = strtok(copy, " ");
    int first_word = 1;

    while (token != NULL) {
        char punct[10] = "";
        int len = strlen(token);
        if (token[len - 1] == '.' || token[len - 1] == ',') {
            punct[0] = token[len - 1];
            punct[1] = '\0';
            token[len - 1] = '\0';
        }

        char word_to_append[LEN];
        strcpy(word_to_append, token);

        for (int t = 0; t < TOKENS; t++) {
            if (strcmp(token, tokens[t]) == 0) {
                int cmd_idx = (t == 0 ? 0 : (t == 1 ? 1 : (t == 2 ? 2 : 3)));
                if (used[cmd_idx] < cl[cmd_idx]) {
                    strcpy(word_to_append, lists[cmd_idx][used[cmd_idx]++]);
                }
                break;
            }
        }

        strcat(word_to_append, punct);
        if (!first_word) strcat(buffer, " ");
        strcat(buffer, word_to_append);
        first_word = 0;

        token = strtok(NULL, " ");
    }
    printf("%s\n", buffer);
}

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return 0;  
}
```

Input:

There once was a [AJ] [N] from [N] who [AV] [V] all day.

NOUNS

squirrel

London

tree

lettuce

ADVERBS

silently

quickly

VERBS

skipped

ran

climbs

ADJECTIVES

delightful

homely

quain

END

Output:

There once was a delightful squirrel from London who silently skipped all day.

There once was a homely tree from lettuce who quickly ran all day.